

MLOnEdge Phase_02 document

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Title	LogiEdge: Intelligent Edge AI Platform for Cold-Chain Logistics
Repository path	https://github.com/2024ac05306-droid/MLOnEdge_miniproject

Task C2 — Preprocessing Pipeline

Path - EdgeAiminiproject-main\EdgeAiminiproject-main\scripts

- Mandatory experiment: run inference with correct stats, then with stats shifted by 3σ ; report accuracy changes in Phase 2 Report

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16
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS E:\MTech\Sem_3\MLOnEdge> C:\Users\hp\miniconda3\envs\logibridge\python.exe e:\MTech\Sem_3\MLOnEdge\scripts\run_ood_experiment.py
2026-08-09 03:16:46.497722: I tensorflow/core/util/port.cc:113] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-08-09 03:16:49.760558: I tensorflow/core/util/port.cc:113] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
=====
MANDATORY EXPERIMENT: OUT-OF-DISTRIBUTION (3σ SHIFT)
=====
Correct Stats Baseline -> Accuracy: 84.29% | Class 2 Recall: 73.08%
+3σ Shifted Stats -> Accuracy: 80.00% | Class 2 Recall: 61.54%
Absolute Accuracy Drop -> 4.29%
=====
PS E:\MTech\Sem_3\MLOnEdge>
```

Interpretation:

When the normalization statistics were deliberately shifted by 3σ , the model received incorrectly normalized inputs. This caused:

- Accuracy to decrease from **84.29% → 80.00%**
- Class 2 recall to decrease from **73.08% → 61.54%**

The **12.54 percentage-point drop in Class 2 recall** is particularly important because it shows that incorrect preprocessing statistics can disproportionately affect detection of Class 2 cases.

The 3σ out-of-distribution experiment demonstrated the sensitivity of the deployed model to normalization statistics. Using the correct training statistics produced 84.29% accuracy and 73.08% Class 2 recall. When the normalization statistics were shifted by 3σ , accuracy decreased to 80.00% and Class 2 recall decreased to 61.54%, resulting in an absolute accuracy drop of 4.29 percentage points. This confirms that consistent versioning and deployment of training-time preprocessing statistics is essential for reliable edge inference.